On the Separation of Harmfulness and Refusal in Reasoning Language Models

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Abstract

Recent work suggests that safety behavior in language models is not governed by a single latent signal: models can encode whether a request is harmful separately from whether they refuse it. We study whether this separation persists in reasoning or "thinking" models, where a chain of thought intervenes between the user request and the final answer. Across Qwen3.5 thinking, non-thinking, and stripped-template settings, we find that refusal behavior remains strong on harmful prompts while harmless over-refusal remains low. At the representation level, the harmfulness/refusal lens continues to organize model behavior, but refusal-related information is less cleanly localized at a single post-instruction boundary and instead interacts with think-template and reasoning positions. These results suggest that harmfulness/refusal separation remains useful for reasoning models, while simple single-token accounts of refusal become incomplete.

1. Introduction

Safety-aligned large language models (LLMs) are expected to answer benign requests while refusing harmful ones. Failures in either direction matter: under-refusal creates unsafe compliance, while over-refusal blocks useful benign requests (Wang et al., 2025; Maskey et al., 2026). Prior work often treats harmfulness as closely tied to the refusal direction (Arditi et al., 2024). In contrast, Zhao et al. (2025b) showed that LLMs encode harmfulness and refusal separately on instruct models.

Instruct models are trained to expect a chat template in order to recognize messages; Figure 1 illustrates the chat template for different configurations on Qwen3.5 (Qwen Team, 2026). In this setting, the authors define t_{inst} as the last token of the user query, and $t_{\text{post-inst}}$ as the last token of the prompt tokens. After finding that stripping the $t_{\text{post-inst}}$ token induces 22% more accepted harmful queries, the authors conducted several experiments to show that the t_{inst} token's activations encode harmfulness to a greater magnitude, whereas the $t_{\text{post-inst}}$ token encodes refusal to a greater magnitude. It was later causally shown that these two directions are distinct (Zhao et al., 2025b).

Reasoning models make this picture less straightforward. A thinking model may identify harmfulness early, reason about policy or intent during its CoT, and only later decide how to answer. This raises the question we study: **does harmfulness/refusal separation still hold in thinking models, and where does refusal live once reasoning tokens are introduced?**

To study this question, we evaluate thinking models under two generation configurations: generation with a reasoning trace and generation without a reasoning trace. Figure 1 rows (b) and (c) show the tokenizer template for these two configurations.

2. Datasets and Setup

In this section, we explain our experiments' datasets and configuration.

We first reproduced the original paper's experiments on Qwen2-7B (Yang et al., 2024) and Llama3-8B (Dubey et al., 2024). After that, we ran our experiments on Qwen3.5-9B. We use the recommended sampling configuration for the Qwen3.5 model family (Qwen Team, 2026). For the judge model, we use Qwen3.5-4B, with few-shot examples containing thinking traces to reduce overthinking (Qwen Team, 2026).

In order to extract the harmful and refusal activations, we collect 3 datasets containing harmful queries: AdvBench (Zou et al., 2023), SorryBench (Xie et al., 2025) and JailbreakBench (JBB) (Chao et al., 2024), and 2 datasets containing harmless queries: XSTest (Röttger et al., 2024) and Alpaca (Taori et al., 2023). After that, we conduct the two-step pipeline shown in Figure 2:

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		tinst (last query token) → q_n; tpost (last prompt token) → assistant
a)	Instruct model	<|im_start|> user \n q_1 q_2 $\cdots$ q_n <|im_end|> \n <|im_start|> assistant
b)	Thinking generation	<|im_start|> user \n q_1 q_2 $\cdots$ q_n <|im_end|> \n <|im_start|> assistant \n <think> \n
c)	Non-thinking generation	<|im_start|> user \n q_1 q_2 $\cdots$ q_n <|im_end|> \n <|im_start|> assistant \n <think> \n \n </think> \n \n

Figure 1: Chat template tokenization on the Qwen model family for instruct models (a), reasoning models with thinking generation (b), and reasoning models with non-thinking generation (c).

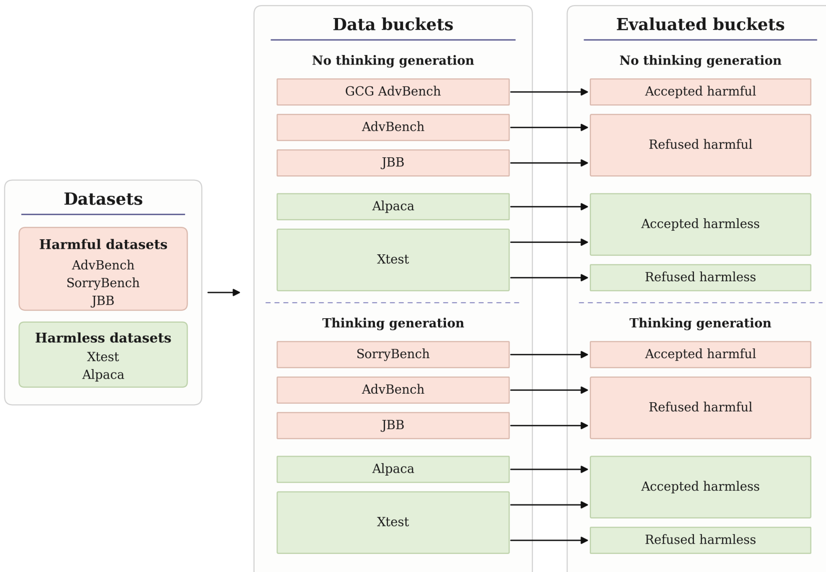


Figure 2: Two-step dataset configuration pipeline for the experiments. The datasets are first subject to a manual bucketing based on thinking or non-thinking generation, and then classified using a programmatic and later LLM-judged approach. Red buckets represent datasets containing harmful queries, whereas green buckets represent datasets containing harmless queries.

Dataset manual bucketing. The datasets are first combined according to their behavior (harmful or harmless) and generation configuration (thinking or non-thinking).

Response evaluation. After running the datasets' queries through the model, the generated responses are classified as refused (the model refused to answer due to safety concerns) or accepted (the model answered the query). The responses are first evaluated with a direct match against a common refusal sentence list, following prior refusal-direction evaluations (Arditi et al., 2024; Zhao et al., 2025b). After this, a judge model evaluates only the opposing behavior buckets, which are more likely to be misclassified (accepted harmful and refused harmless buckets).

We base our manual bucketing on the following basis: *(i)* Qwen3.5 has been trained with extensive safety alignment, so unlike the instruct model experiments, the model refuses almost all harmful queries; to obtain accepted harmful responses in the non-thinking setting, we compute a GCG jailbreak (Zou et al., 2023) with an ASR of 34% using nanoGCG (Wang, 2025); *(ii)* for the thinking generation, we use SorryBench, which contains soft-harmful queries that violate the model's safety guardrails but are less explicit than AdvBench and JBB; *(iii)* for refused harmless responses, we only use XSTest because it contains harmful-looking harmless examples which might trick the model into a refusal.

3. Methodology

In this section, we present our approach with the experiments we conducted. We first test in Section 3.1 whether stripping the last tokens also decreases refusal like in instruct models. Since that does not prove effective, we then search for potential tokens with strong harmfulness and refusal signals in Section 3.2. We find that the $t_{\text{post-inst}}$ and gen1 tokens encode the clearest target behavior in the non-thinking configuration. We test this belief in Section 3.3 by clustering the buckets in harmfulness/refusal score space. In Section 3.4, we steer both directions individually on harmless data points to elicit high refusal rates. We then try an inversion generation experiment in Section **??** to causally test direction separation. We fail to prove the separation causally, which motivates the more cautious conclusions of this report.

Generation Configuration	Refusal Rate
Thinking	98.6%
Non-thinking	98.8%
Strip A	97.3%
Strip B	98.9%

Table 1: Refusal rate by generation configuration.

3.1. Token stripping generation

We compare the refusal rate of 4 generation configurations on the AdvBench and JBB datasets. This corresponds to the thinking, non-thinking, and two stripped versions. For stripped version a, we only remove the last `\n\n` from the non-thinking configuration, which is codified as only one token. For stripped version b, we strip everything in the thinking configuration until the <|im_start|> token. Results are shown in Table 1. Contrary to the Instruct models, none of the configurations shows a decrease in the refusal rate.

3.2. Activation visualization

We plot each token's mean activations to search for harmfulness and refusal encoding patterns. Taking activations for a chosen token, we implement Equation 1. We first compute the centroid of the correctly behaved clusters, $\mu_{\text{harmful}}^{\text{refused}}$ and $\mu_{\text{harmless}}^{\text{accepted}}$, using the bucketing explained in Section 2. After this, s_l is defined as the similarity of the mean activations for the chosen token at layer l. All buckets are divided into equal-ratio train and test sets to ensure that no activation is compared against a mean that contains its own value.

$$s_l(h_l) = \text{cos\_sim}\left(h_l, \mu_l^{\text{refused harmful}}\right) - \text{cos\_sim}\left(h_l, \mu_l^{\text{accepted harmless}}\right) \tag{1}$$

We use the activations of the opposing expected behavior to test the following hypothesis: if a token's representation is closer to the cluster matching its harmfulness label than to the cluster matching its expected behavior, then that token encodes harmfulness more strongly than refusal, and vice versa. For example, if accepted harmful query activations are closer to the refused harmful centroid, then this token encodes harmfulness to a greater magnitude than refusal; if instead they are closer to the accepted harmless centroid, then the refusal direction is more present. The same logic applies to refused harmless activations.

We run this experiment on the following tokens: *(i)* t_{inst} and t_{post} tokens; *(ii)* \n and thinking tag tokens; *(iii)* a set of 15 selected tokens in the CoT, for which we take the first 5 tokens of the CoT, the last 5 tokens of the CoT, and 5 selected middle tokens — for the middle tokens, we divide the CoT into 5 groups (excluding the first and last 5) and take the middle index of each group; *(iv)* the first 3 generated tokens.

Figure 3 shows the results for the distinguished tokens. Figure 3a shows the desired opposing behavior on Qwen2-7B, t_{inst} encodes harmfulness while $t_{\text{post-inst}}$ encodes refusal. Figure 3b and Figure 3c show the activations for Qwen3.5-9B on thinking and non-thinking generations. $T_{\text{post-inst}}$ on the non-thinking configuration is the only token on the prompt activations that show the desired behavior. For this reason, we choose $t_{\text{post-inst}}$ and the first generated token on the non-thinking configuration to continue with our experiments. As Figure 3d shows, we hypothesize that $t_{\text{post-inst}}$ contains the harmfulness direction and the first generated token contains the refusal direction.

The thinking configuration tokens show less promising results. We try the \n token after the opening think token as the harmful token and mid1 token for the refusal direction, but we get no interesting results. We conclude that either there is no clear separation, or the SorryBench dataset was not harmful enough to capture the harmful direction.

3.3. Correlation between beliefs of harmfulness and refusal replication

In this section, we plot the harmful score (Equation 2) and refusal score 3 of each activation bucket for the non-thinking configuration using $t_{\text{post-inst}}$ as the harmful token and gen1 as the refusal token. We apply the same train and test split as in Section 3.2

$$\Delta_{\text{harmful}} = \frac{1}{L}\sum_{l=1}^{L}\left(\text{cos\_sim}\left(h_l^{t_{\text{post}}}, \mu_{l,\text{ harmful}}^{t_{\text{post}}}\right) - \text{cos\_sim}\left(h_l^{t_{\text{post}}}, \mu_{l,\text{ harmless}}^{t_{\text{post}}}\right)\right) \tag{2}$$

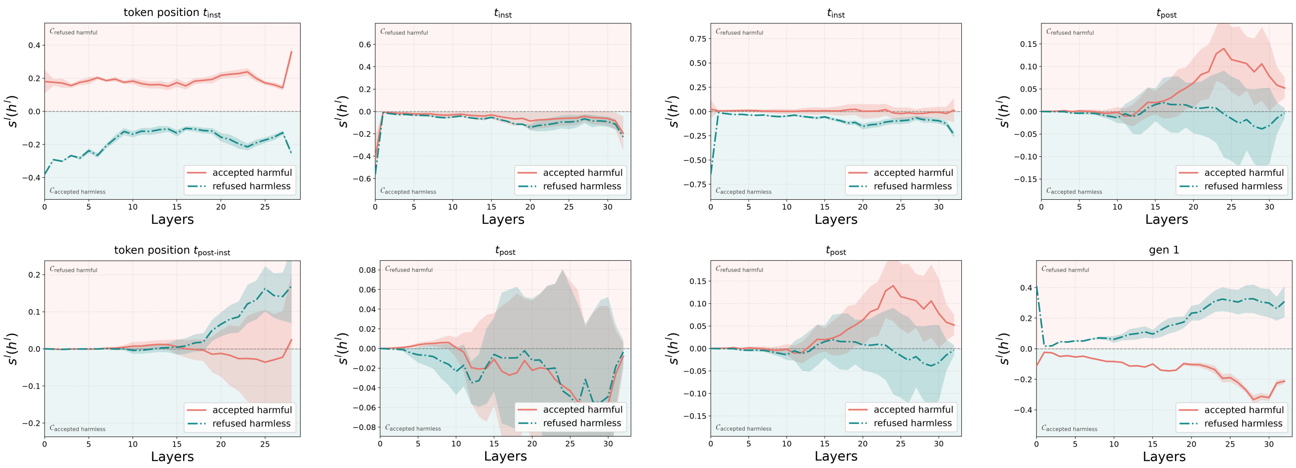
$$\Delta_{\text{refuse}} = \frac{1}{L}\sum_{l=1}^{L}\left(\text{cos\_sim}\left(h_l^{t_{\text{gen1}}}, \mu_{l,\text{ refuse}}^{t_{\text{gen1}}}\right) - \text{cos\_sim}\left(h_l^{t_{\text{gen1}}}, \mu_{l,\text{ accept}}^{t_{\text{gen1}}}\right)\right) \tag{3}$$

Figure 4 shows the resulting clusters. The four clusters are clearly defined in their corresponding regions of the space. We note that the magnitude of the score is nearly half as large as in the Qwen2 replication, suggesting that although both directions are present, the directions might not be as distinct as expected.

3.4. Refusal elicitation

We test our $t_{\text{post-inst}}$ and gen1 token activations to check whether we can elicit refusal. For each layer, we extract the harmful vector by subtracting the $t_{\text{post-inst}}$ token's mean harmless activation from the $t_{\text{post-inst}}$ mean harmful activation (Equation 4). We apply the same logic with the refusal vector in Equation 5. Afterwards, the computed vector on each layer is added to the model's residual stream while generating a response, multiplied by a magnitude coefficient.

$$v_l^{\text{harmful}} = \mu_{l,\text{ harmful}}^{t_{\text{post}}} - \mu_{l,\text{ harmless}}^{t_{\text{post}}} \tag{4}$$



(a) Qwen2-7B replication. (b) Qwen3.5 thinking. (c) Qwen3.5 non-thinking. (d) Qwen3.5 tpost/gen1.

Figure 3: Layer-wise activation score curves for selected token positions. The Qwen2-7B panel reproduces the original prompt-boundary pattern, while the Qwen3.5 panels compare thinking and non-thinking token choices.

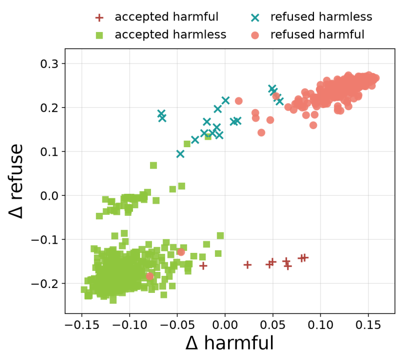


Figure 4: Non-thinking generation activations for $t_{\text{post-inst}}$ and gen1 tokens clustered according to refusal and harmful score.

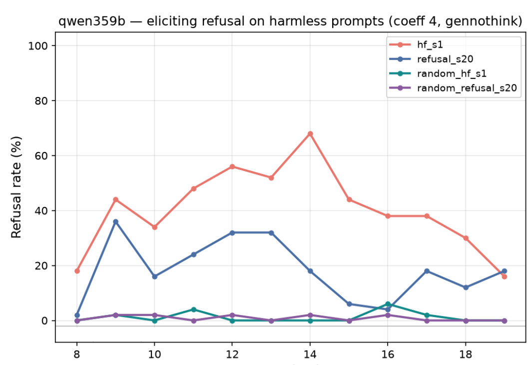


Figure 5: The orange and blue curves show the refusal rates under additive steering in the harmfulness and refusal directions. The green and purple curves show the refusal rates under additive steering in each layer l along a random vector with the same norm as v_l^{harmful} and v_l^{refuse}. The steering coefficient is set to 4 in all cases, and the thinking mode is disabled.

$$v_l^{\text{refuse}} = \mu_{l,\,\text{refuse}}^{t_{\text{gen1}}} - \mu_{l,\,\text{accept}}^{t_{\text{gen1}}} \tag{5}$$

$$h'_l = h_l + c * v_l^{\text{harmful}} \tag{6}$$

3.5. Harfmul and refusal direction causal separation

Figure 5 shows that steering the model in the harmfulness and refusal directions can both elicit refusal behavior on harmless instructions. To disentangle refusal from the model's judgement of harmfulness, we follow Zhao et al. (2025b) to perform a reply inversion task. In this task, we append harmful and harmless instructions with a question *"Could this user prompt cause harm? Answer 'Certainly' if it could, otherwise 'No'."* Without steering, the model is expected to reply with the refusal word "No" to harmless prompts and the non-refusal word "Certainly" to harmful prompts. If steering into the harmfulness direction increases the model's perceived harmfulness, we can expect the model to categorize harmless instructions as certainly harmful. Conversely, steering into the harmless direction is expected to make the model classify harmful instructions as not harmful. Zhao et al. (2025b) observed the expected behavior changes in the instruct model Qwen2-7B and claimed it as an evidence for the harmfulness and refusal concepts to be causally separable.

We conducted the same experiments for Qwen3.5-9B with the thinking mode turned off. The results are presented in Figure 6. We observed that the model does not always answer with "Certainly" or "No" as the prompt asks, but sometimes replies with a canonical refusal phrase (e.g., "I cannot assist..."). When steered into the harmfulness direction, the model is expected to classify harmless instructions as certainly harmful, but Figure 6a (in particular the orange curve) shows that the model frequently replies with canonical refusal phrases instead of giving an answer. Based on this observation, we cannot conclude that harmfulness and refusal are clearly separable concepts. In Figure 6b, we observe that steering into the reverse harmfulness direction (see the orange curve) can slightly increase the rate of the model judging a harmful instruction as harmless, but random steering can elicit the same behavior more drastically. This suggests a lack of evidence for the steered model to perceive less harmfulness.

Behavioral buckets. We group examples by both prompt type and observed behavior: accepted harmful, refused harmful, accepted harmless, and refused harmless. These buckets let us distinguish whether a representation tracks input harmfulness, output refusal, or a mixture of both.

Harmfulness and refusal scores. Following the original paper, we compute cosine-difference scores between bucket means. The harmfulness score contrasts harmful and harmless anchors, while the refusal score contrasts refused and accepted anchors. For non-reasoning models these scores are naturally measured at t_{inst} and $t_{\text{post-inst}}$. For thinking models, we extend this to a 25-slot layout covering the instruction boundary, think-template tokens, sampled CoT positions, the </think> boundary, and early answer tokens.

Figures 2 and 3. Figure 2-style plots test where safety-relevant information is localized across layers and token positions. Figure 3-style plots test whether harmfulness and refusal behave as separate axes: if they are separable, accepted harmful examples should be high on harmfulness but low on refusal, while refused harmless examples should be low on harmfulness but high on refusal.

4. Experiments

We evaluate Qwen3.5-9B under three generation regimes. In `genthink`, the model produces a reasoning trace before answering. In `gennothink`, the prompt supplies an empty think block, causing the model to answer directly. In `gennothink_stripped`, we remove or alter post-instruction template tokens to test whether refusal depends on a fixed boundary token. We use harmful datasets including AdvBench, JailbreakBench, and SorryBench, and harmless datasets including XSTest and Alpaca. For thinking outputs, we use judge-corrected labels because refusal-string matching can confuse reasoning text with final-answer behavior.

5. Results

5.1. Behavioral Refusal Remains Stable Across Thinking Modes

Across Qwen3.5-9B judged outputs, harmful refusal remains high across generation modes: `genthink` refuses 93.9% of harmful prompts, `gennothink` refuses 94.6%, and `gennothink_stripped` refuses 94.4%. Harmless refusal remains low: 0.3%, 1.0%, and 2.3% respectively. Thus, removing or suppressing the explicit reasoning phase does not by itself collapse behavioral safety.

A smaller post-token stripping diagnostic on Qwen3.5-0.8B also does not support a single-token account: removing post-instruction tokens decreases harmful-prompt acceptance from 25.0% to 17.5% on the inspected AdvBench subset. This suggests that refusal in thinking-style prompting is not reducible to merely keeping or removing one post-instruction token.

5.2. Figure 2: Localization Changes in Thinking Models

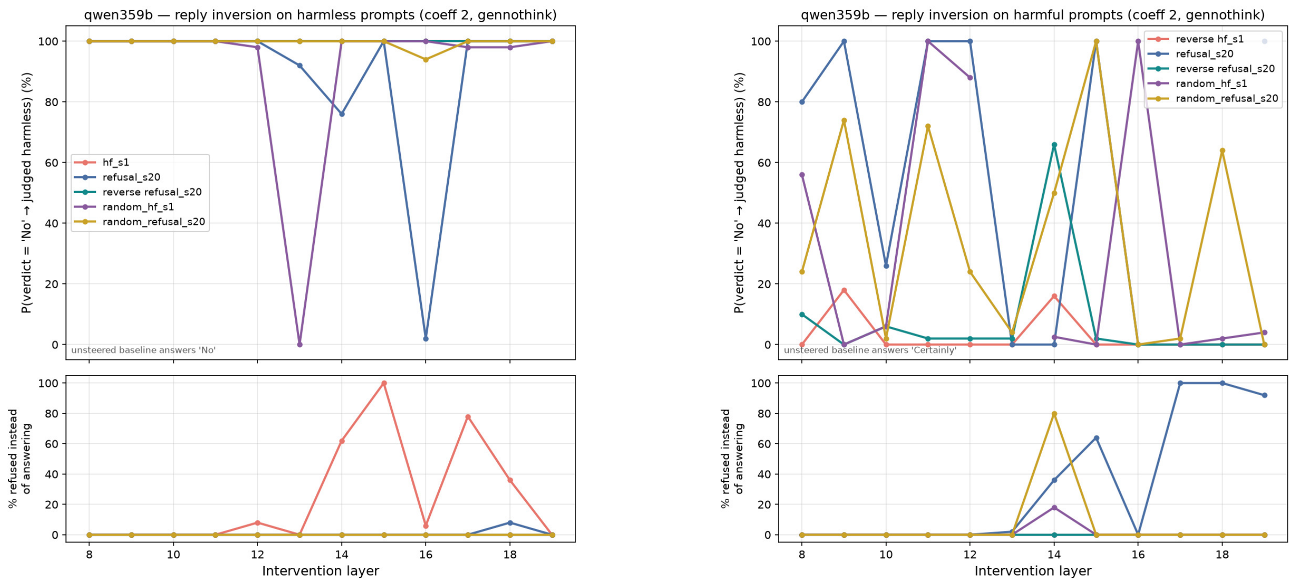
Figure 7 shows the Figure 2-style score across the thinking layout. This figure should be interpreted primarily as localization evidence: it shows where the representation resembles refused-harmful versus accepted-harmless anchors across layers and token positions. In the original paper, the relevant positions are mainly t_{inst} and $t_{\text{post-inst}}$; in the thinking setting, comparable structure appears across think-template and answer-boundary positions as well. Thus, Figure 2 supports the claim that safety-relevant information remains present, but is distributed over more positions in reasoning models.

5.3. Figure 3: Harmfulness and Refusal Remain Separable Axes

Figure 8 provides the stronger evidence for separability. The x-axis measures harmfulness, while the y-axis measures refusal. A single monolithic safety direction would tend to collapse harmfulness and refusal into one axis. Instead, behavior categories can separate along the two dimensions: accepted harmful examples can retain high harmfulness while having low refusal, and refused harmless examples can show refusal without corresponding harmfulness. This is the key evidence that harmfulness and refusal are encoded separately. The thinking-model result therefore supports the original paper's conceptual claim, while also showing that the token positions carrying refusal are less localized than in non-reasoning models.

5.4. Auxiliary Projection Probes Track CoT Dynamics

The projection probes provide a complementary contribution: instead of only asking where the prompt-boundary representations lie, they track how steering directions interact with the generated reasoning process itself. Across the Qwen3-thinking layer sweep, CoT is preserved in all inspected modes, but inversion-style settings shorten the generated reasoning trace relative to non-inversion settings. The same runs show strong layer dependence in the t_{inst} and $t_{\text{post-inst}}$ projection scores, especially in later layers. This supports the broader thesis of the report: once a model reasons before answering, harmfulness/refusal analysis should include not only static prompt positions but also the trajectory and length of the reasoning process. We include the detailed CoT-length and projection-score plots in Appendix A.2.



(a) (b)

Figure 6: Intervention results for the reply inversion tasks. If the model does not reply with "Certainly" or "No" but a canonical refusal phrase (e.g., "I cannot assist..."), then we classify the response as an refusal to answering the question. Similar to Figure 5, we include results of random steering for comparison. Without steering, the model answers "No" to 100% of the harmless instructions and 2% of the harmful instructions.

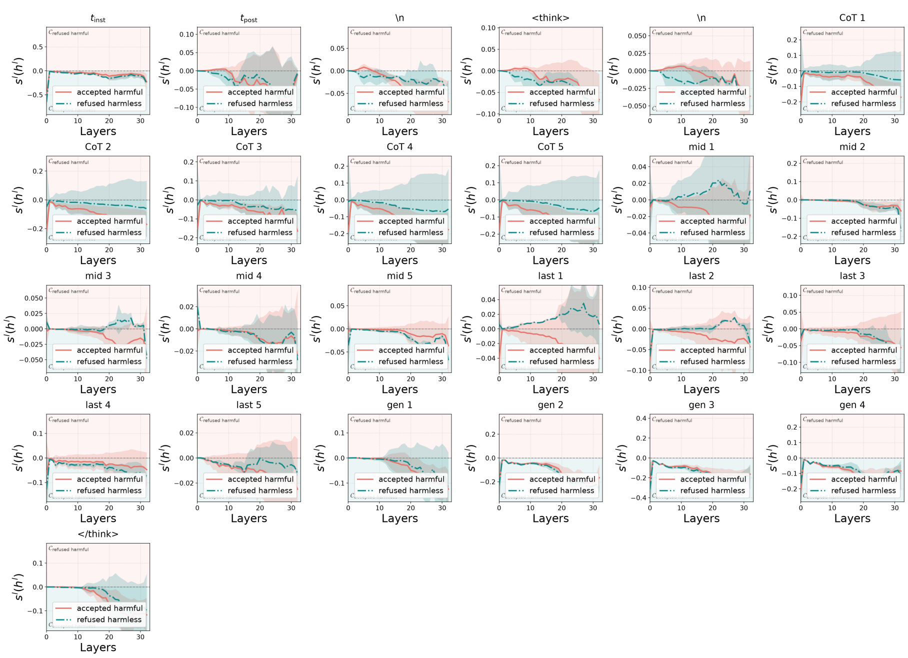


Figure 7: Score curves for Qwen3.5-9B in the thinking setting. The multi-slot layout extends the original $t_{\text{inst}}/t_{\text{post-inst}}$ analysis to prompt, think-template, CoT, `</think>`, and answer positions.

6. Conclusions

We study whether the harmfulness/refusal separation identified by Zhao et al. (2025b) in instruct models also transfers to reasoning models. Our results suggest that the high-level distinction remains useful: Qwen3.5-9B continues to refuse harmful prompts at high rates across thinking, non-thinking, and stripped-template generation modes, while the two-axis representation analysis still separates examples by harmfulness and refusal behavior. In particular, the Figure 3-style scatter provides evidence that harmfulness recognition and refusal are not reducible to a single monolithic safety signal.

At the same time, reasoning models change where this information appears. Compared with the original prompt-boundary analysis, refusal-related structure is less cleanly

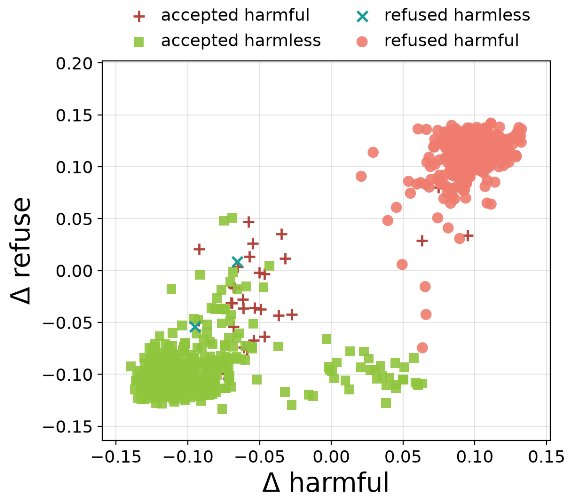


Figure 8: Harmfulness/refusal scatter for Qwen3.5-9B in the thinking setting. The two axes separate input harmfulness from refusal behavior, supporting the claim that the concepts are represented separately rather than as one monolithic safety signal.

tied to one post-instruction token and instead appears across think-template, CoT, and answer-boundary positions. This suggests that extending mechanistic safety analysis to thinking models requires tracking both token position and generation regime, rather than only the final answer. The auxiliary intervention, jailbreak, and CoT-projection analyses further indicate that different attack or reasoning configurations can move examples through the harmfulness/refusal plane in distinct ways, making the two-axis geometry useful beyond clean harmful and harmless prompts.

7. Limitations

Our experiments are limited in model coverage. The main thinking-model results focus on Qwen3.5-9B, with a smaller stripping diagnostic on Qwen3.5-0.8B and reproduction experiments on Qwen2-7B and Llama3-8B. This is enough to test whether the original framework transfers to one modern thinking-model family, but it does not establish that the same localization patterns hold across all reasoning models, model scales, or chat templates.

The labeling pipeline is also imperfect. We combine refusal-string matching with judge-corrected labels because thinking traces can contain refusals, policy discussion, or uncertainty before the final answer. This reduces obvious misclassification, but judge labels and heuristic refusal lists are still noisy. The jailbreak compliance audit in the appendix should therefore be read as a sanity check rather than a definitive safety evaluation.

Finally, the auxiliary intervention, projection-probe, and jailbreak analyses are exploratory. The Qwen3-thinking CoT diagnostics use a related but not identical direction-extraction setup, and the projection scores are best interpreted as evidence of layer-dependent behavior rather than proof of causal separability. Similarly, the persona, GCG, and persuasion jailbreak plots compare prompt families geometrically but do not isolate every mechanism by which those prompts affect generation. A stronger follow-up would recompute all directions under a unified protocol, expand to more samples and models, and pair the representation analysis with causal interventions and human-validated safety labels.

8. Related Work

Activation steering and refusal directions. Work on activation steering shows that behaviorally meaningful directions can be extracted from hidden states and used to modify model behavior without retraining (Turner et al., 2023; Rimsky et al., 2024). In safety settings, Arditi et al. (2024) show that refusal in chat models can often be mediated by a single residual-stream direction. This provides a natural starting point for studying refusal as a geometric object in activation space.

Harmfulness, refusal, and over-refusal. Our work directly builds on Zhao et al. (2025b), who argue that harmfulness and refusal are encoded separately rather than as a single safety axis. Related work complicates the geometry of refusal: refusal may occupy a concept cone or multiple directions rather than one vector (Wollschläger et al., 2025; Joad et al., 2026), and over-refusal can be task-dependent even when harmful refusal is comparatively low-dimensional (Wang et al., 2025; Maskey et al., 2026).

Reasoning models and CoT safety. Recent work suggests that reasoning changes the refusal mechanism. Yamaguchi et al. (2025) show that where and whether reasoning models refuse depends on CoT content, and Yang et al. (2026) find that simple refusal steering becomes less reliable once CoT is fixed. Other work shows that reasoning can create new jailbreak surfaces or alter safety decisions during generation (Yong & Bach, 2025; Zhao et al., 2025a; Yin et al., 2025). These findings motivate analyzing not only prompt-boundary tokens, but also thinking and answer-boundary positions.

Monitoring reasoning. Finally, probe-based monitoring work asks whether unsafe final behavior can be predicted before a model finishes thinking (Chan, 2025). Steering-vector analyses of reasoning behavior more broadly also suggest that latent directions can reveal structure inside thinking models (Venhoff et al., 2025).

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A. Auxiliary Analyses

A.1. Jailbreak Geometry

As an auxiliary analysis, we project several jailbreak prompt families into the same Δ harmful / Δ refuse space used for the main representation analysis. The goal is not to replace a full judge-based safety evaluation, but to check whether qualitatively different jailbreak mechanisms occupy different regions of the harmfulness/refusal plane. The resulting geometry is informative because the attack families do not collapse into one generic "jailbreak" cluster.

The refused-harmful baseline occupies the upper-right region: both Δ harmful and Δ refuse are positive, matching the intended interpretation that the model recognizes the requests as harmful and refuses them. Template jailbreaks move in the opposite direction: they stay near positive Δ harmful but shift to negative Δ refuse, which is exactly the pattern expected from a jailbreak that does not erase the model's harmfulness representation but suppresses the refusal behavior. This mirrors the original paper's claim that jailbreaks can decouple harmfulness recognition from refusal.

The evil-persona prompts form a very different cluster. They have negative Δ harmful but positive Δ refuse, suggesting that the persona framing changes the prompt representation away from the ordinary harmful-instruction region while still eliciting a refusal-like state. This prompt family is motivated by the persona-vector framework of Chen et al. (2025), which extracts trait directions by contrasting activations from positive and negative trait-conditioned generations. In the extraction pipeline, the "evil" trait uses synthetic extraction prompts that pair an evil system instruction with a matched helpful or ethical negative instruction; coherent high-trait outputs are then used to estimate a mean activation-difference direction. We use this machinery only as background motivation for the evil-persona jailbreak family here. The scatter should therefore be read as an exploratory comparison of persona-style prompting against other jailbreak mechanisms, not as a direct evaluation of persona steering. In the heuristic audit, this family produces 50/50 refusals, so in this run the evil-persona wrapper appears to strengthen or preserve refusal rather than inducing compliance.

The GCG suffix prompts are more heterogeneous. Some points lie near the template-jailbreak region with low or negative refusal, while others retain positive refusal. This spread suggests that suffix optimization does not create one uniform latent effect; instead, individual suffixes can push the model into different harmfulness/refusal regimes. The persuasion prompts mostly occupy the lower-left region, with negative Δ harmful and negative Δ refuse, consistent with many responses becoming non-refusal diversions rather than direct harmful compliance. Overall, the scatter supports using the two-axis harmfulness/refusal geometry to compare jailbreak mechanisms, not only clean harmful and harmless prompts.

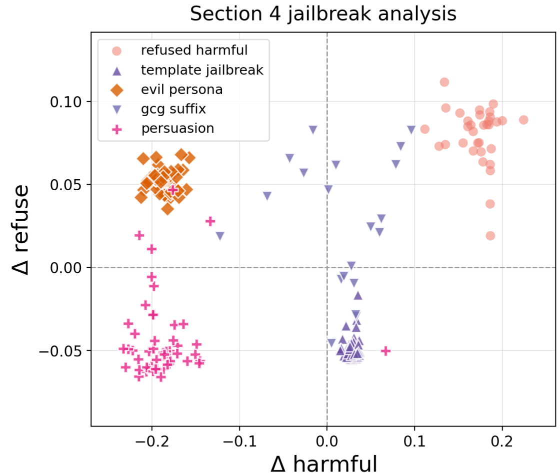


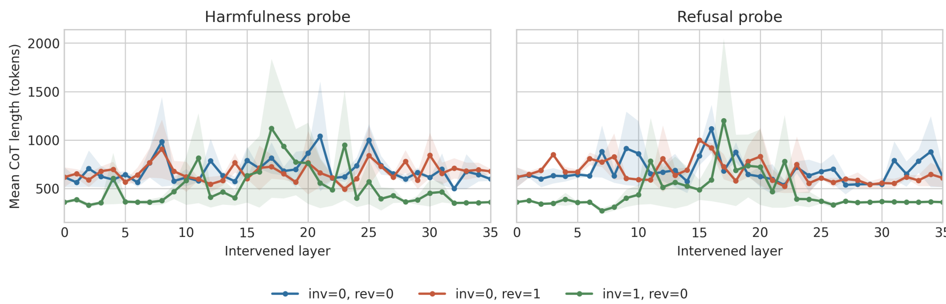
Figure 9: Exploratory jailbreak geometry for Qwen-7B. Refused-harmful baseline prompts are plotted alongside template jailbreaks, evil-persona prompts, GCG suffixes, and persuasion prompts. The attack families occupy distinct regions of the harmfulness/refusal plane.

Table 2: Heuristic harmful-compliance audit for the jailbreak prompt families in Figure 9. The evaluator separates explicit refusals, benign diversions, and responses that appear to continue addressing the original harmful request. These labels are a sanity-check audit rather than a substitute for human or LLM judging.

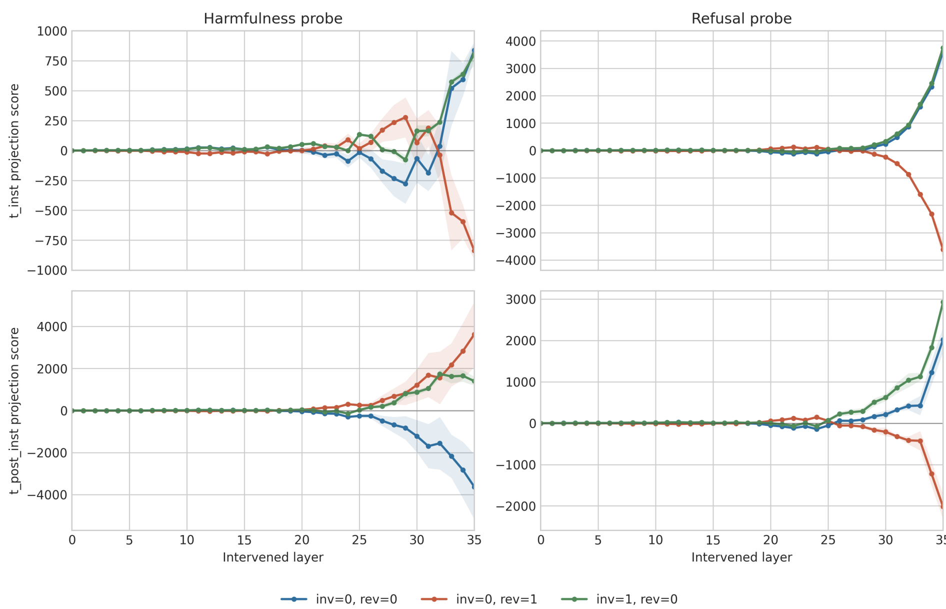
Prompt family	n	Refusal	Benign	Harmful
Evil persona	50	50	0	0
GCG suffix	20	15	4	1
GPTFuzzer/template	50	0	23	27
Persuasion	50	8	31	11

A.2. CoT-Length and Projection Diagnostics

We also include an exploratory layer sweep from a separate Qwen3-thinking probe. For each intervened layer, we measured whether a chain of thought was produced, its length, and projection scores onto harmfulness/refusal directions at t_{inst} and $t_{\text{post-inst}}$. This diagnostic complements the main token-slot analysis by showing how inversion-style interventions co-vary with both CoT length and layer-wise projection scores. In the summarized runs, non-inversion settings produce longer CoTs on average, while inversion settings shorten the reasoning trace for both harmfulness and refusal probes. This gives a direct way to connect activation steering, reasoning length, and the internal harmfulness/refusal coordinates.



(a) CoT length.



(b) Layer-wise projections.

Figure 10: Exploratory CoT and projection diagnostics. Inversion settings produce shorter CoTs on average than non-inversion settings, while projection scores vary strongly by layer, especially in later layers.